Yugma TechFest 2.0 - MedhaDrishti National-Level AI Hackathon

Topic: AI for Medical Image Enhancement and segmentation Hackathon Challenge

Theme: Designing Efficient Framework (using ML/DL based AI models) for Magnetic resonance image (MRI volume) enhancement and region of interest segmentation.

Background:

MRI enhancement and diseased tissue/organ region segmentation are two fundamental steps in modern medical image analysis. Together, they improve image quality and enable accurate localization and quantification of pathological tissues, supporting both radiologists and AI-based clinical decision support systems. MRI enhancement aims to improve image quality without altering the anatomical or pathological information (NifTI/Nii/Dicom file formats). Recent advances in leveraging AI for MRI enhancement and region of interest (ROI) segmentation use Self-supervised, Semi-supervised and Unsupervised learning models. The region of interest (ROI) may include tumours, stroke infarcts, multiple sclerosis lesions, edema in Brain MRI and intervertebral disc degeneration, disc herniation, spinal stenosis in Spine (LS) MRI. This hackathon aims to develop an efficient Framework for Magnetic resonance image (MRI volume) enhancement and region of interest (ROI) segmentation, which will be evaluated for its application in clinical decision support systems by an expert Radiologist.

Problem Statement

Develop an AI based efficient framework for Magnetic resonance image (MRI volume) enhancement and region of interest segmentation.

Participants may explore the following tasks:

- Curated MRI Dataset Preparation.
- Development of Semi-supervised /Ensemble ML/ Self-Supervised /Unsupervised AI model for Brain MRI (T1, T2, FLAIR) and Spine MRI (T1, T2, STIR) enhancement and ROI segmentation.
- Evaluation of Raw/Enhanced MRI samples with standard metrics of enhancement.
- Evaluation of segmented MRI samples with standard metrics.

Dataset

Brain MRI	**Spine MRI**
Dataset folder size: 445 MB	Dataset folder size: 488 MB
10 no. of normal MRI samples, each sample folder includes T1, Contrast T1(T1c), T2 weighted, FLAIR nii/GZ files.	10 no. of normal MRI samples, each sample folder includes T1, Contrast T1(T1c), T2 weighted, STIR nii/GZ files

10 no. of pathological MRI samples, each sample folder includes T1, Contrast T1(T1c), T2 weighted, FLAIR nii/GZ files	10 no. of pathological MRI samples, each sample folder includes T1, T2 weighted, STIR nii/GZ files

Participants will be provided Dataset offline by Hackathon organizers, containing folders of normal and pathological **Brain, Spinal MRI files in NifTi (GZ/Nii) format**. The dataset folders will contain raw MRI(Nii/GZ) volume samples of normal individual/patients, without any ground truth annotations of region of interest.

IMPORTANT NOTE:

- ***Participants must use MRI dataset samples in NifTI (GZ/Nii) file formats only, at any stage of Hackathon workflow.***
- ***Participants must use Brain MRI standard training dataset given in below link for Brain MRI related tasks.***
  Weblink of Brain MRI Dataset (Standard training Dataset):
  (BRATS-CHALLENGE 2020 Dataset)
  https://www.kaggle.com/datasets/awsaf49/brats20-dataset-training-validation

- ***For LS Spine MRI related tasks, participants must use offline shared Hackathon Dataset only (No external dataset allowed for Spine MRI).***

https://drive.google.com/drive/folders/19psnKwO4swOQ6BUE2xuPpxu0PyJ3SKM1

- ***Participants may discuss with Hackathon volunteers, organisers for any queries/ clarifications regarding dataset/Hackathon challenge workflow.***

Hackathon Workflow:

The hackathon challenge workflow is as per the following stages.

STAGE 1 (Dataset Exploration, Analysis, and Preparation)

Team participants are required to understand and analyse the various MRI anatomical details described in below figures and correlate them with available MRI samples in the Hackathon dataset. Perform Dataset statistics analysis and Brain / Spine MRI Sub-modalities division of raw dataset.

NOTE:

The following figures demonstrate the differences between various sub modalities (T1, T2, FLAIR) of MRI acquisition of human brain and spine.

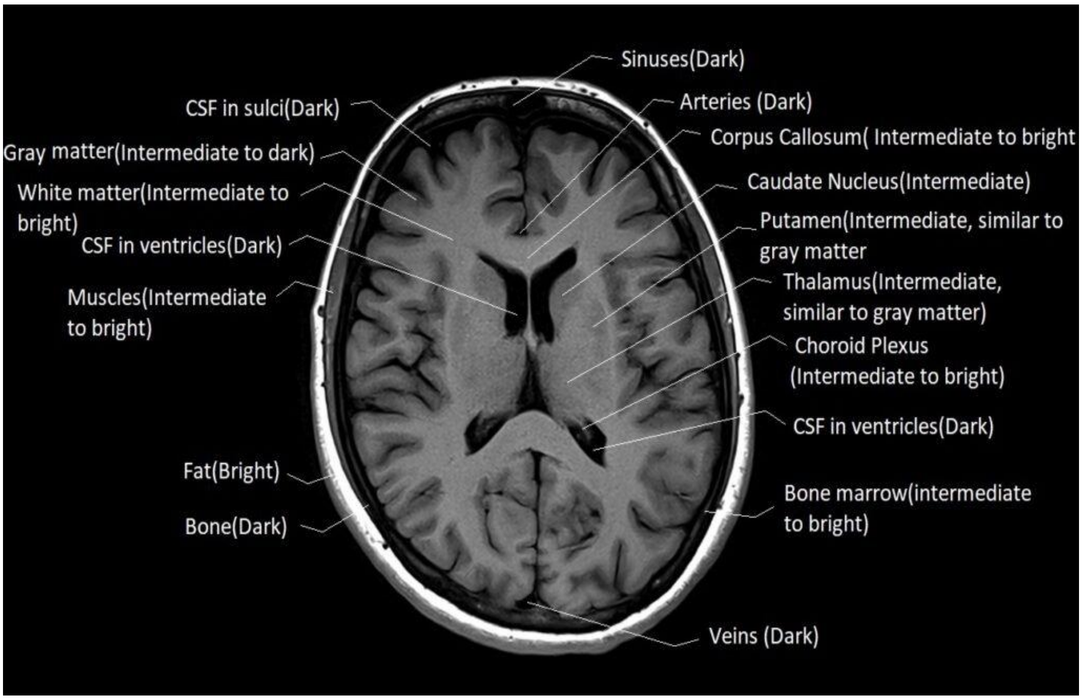


Fig 1: T1 MR Image of the Brain (axial plane)

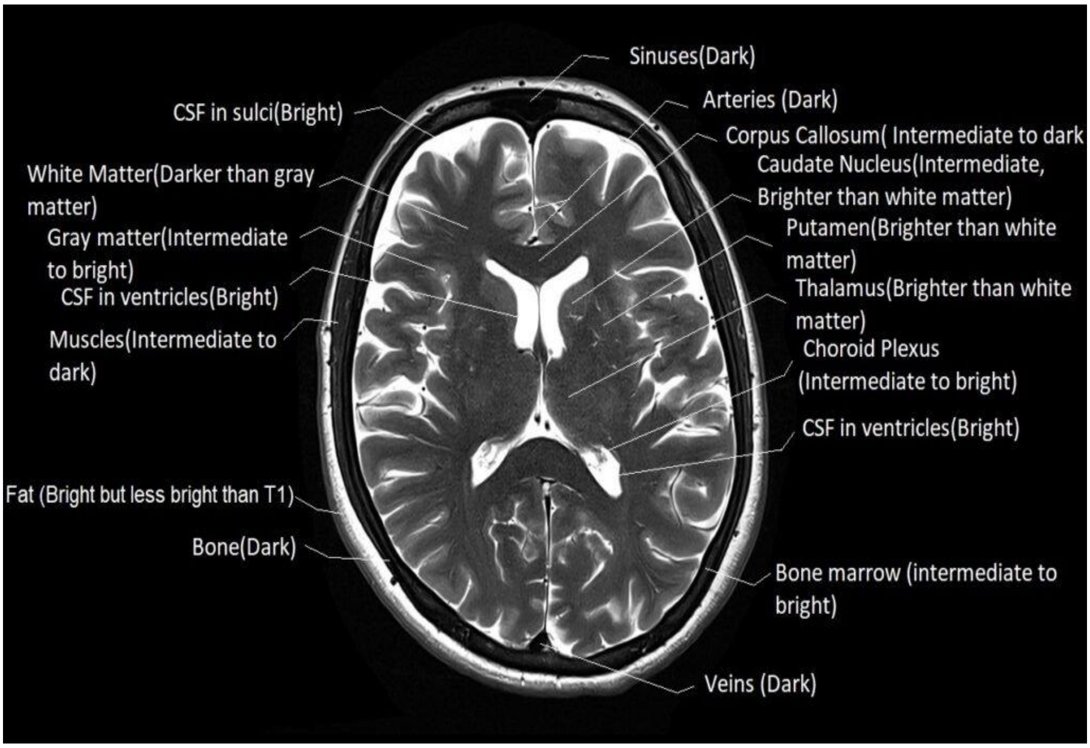


Fig 2: T2 MR Image of the Brain (axial plane)

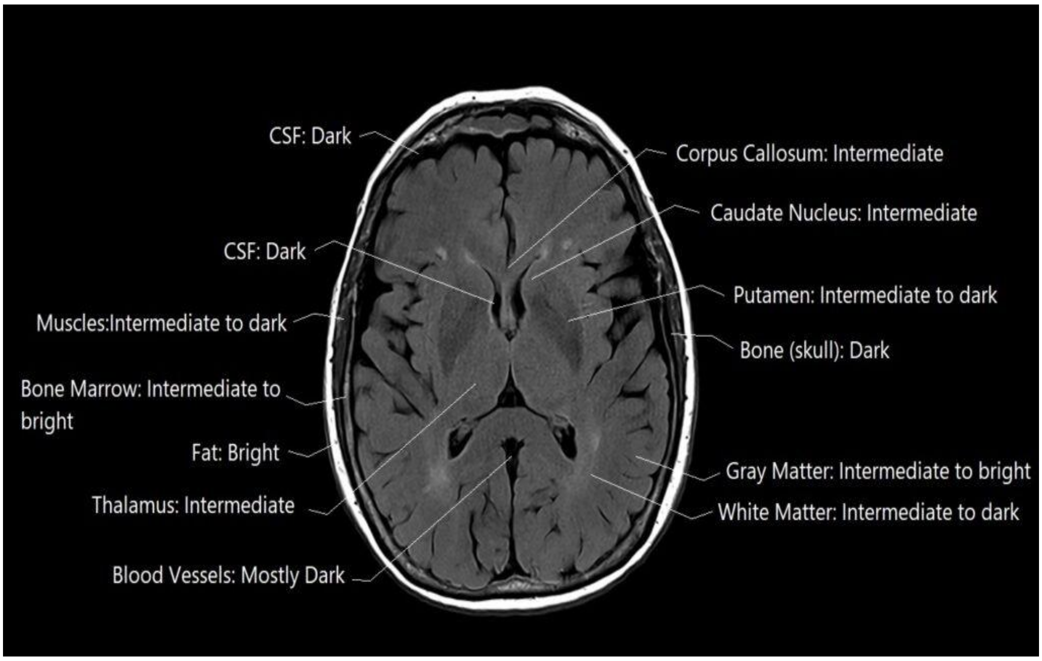


Fig 3: FLAIR MR Image of the Brain (axial plane)

A Lumbar Spine (LS) MRI is a non-invasive imaging examination that uses a strong magnetic field and radiofrequency waves to produce detailed images of the lower part of the spine (lumbar region), surrounding soft tissues, spinal cord, nerve roots, intervertebral discs, muscles, ligaments, and blood vessels. Unlike CT or X-rays, MRI does not use ionizing radiation and provides excellent soft tissue contrast, making it the preferred modality for evaluating spinal disorders.

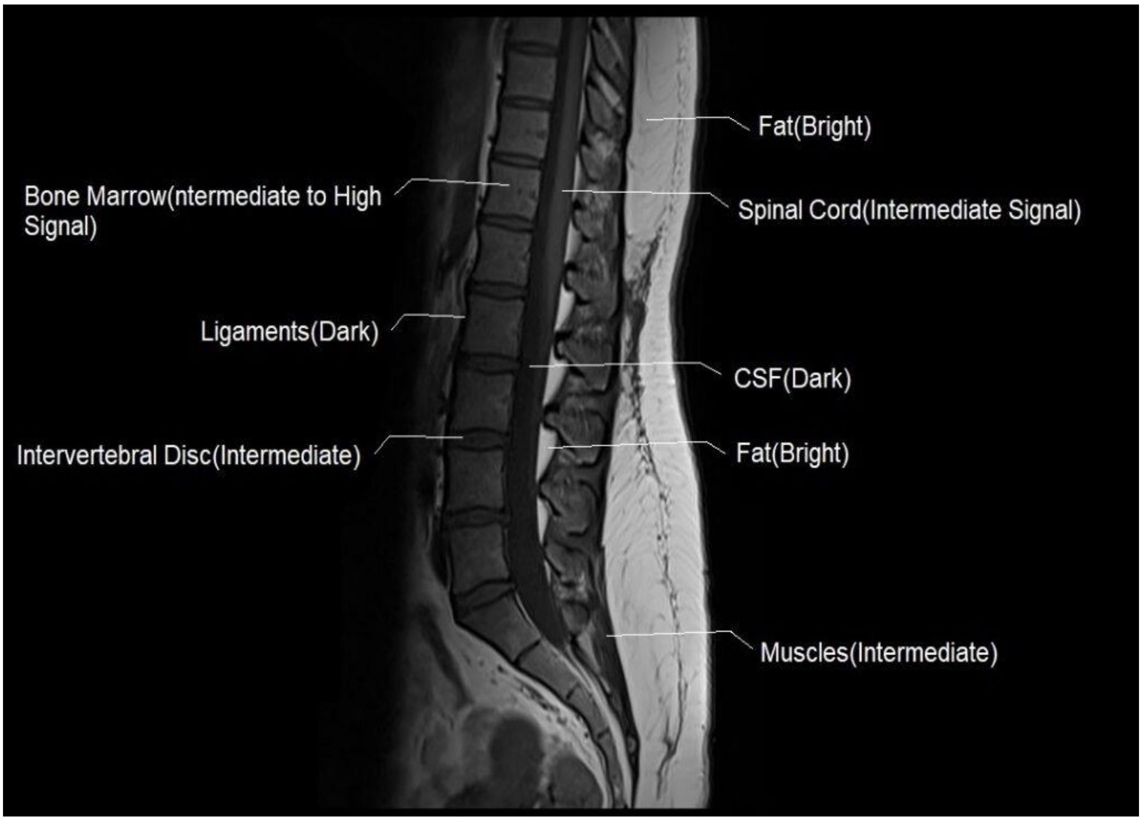


Fig 4: T1 MR Image of the lumbar spine (sagittal plane)

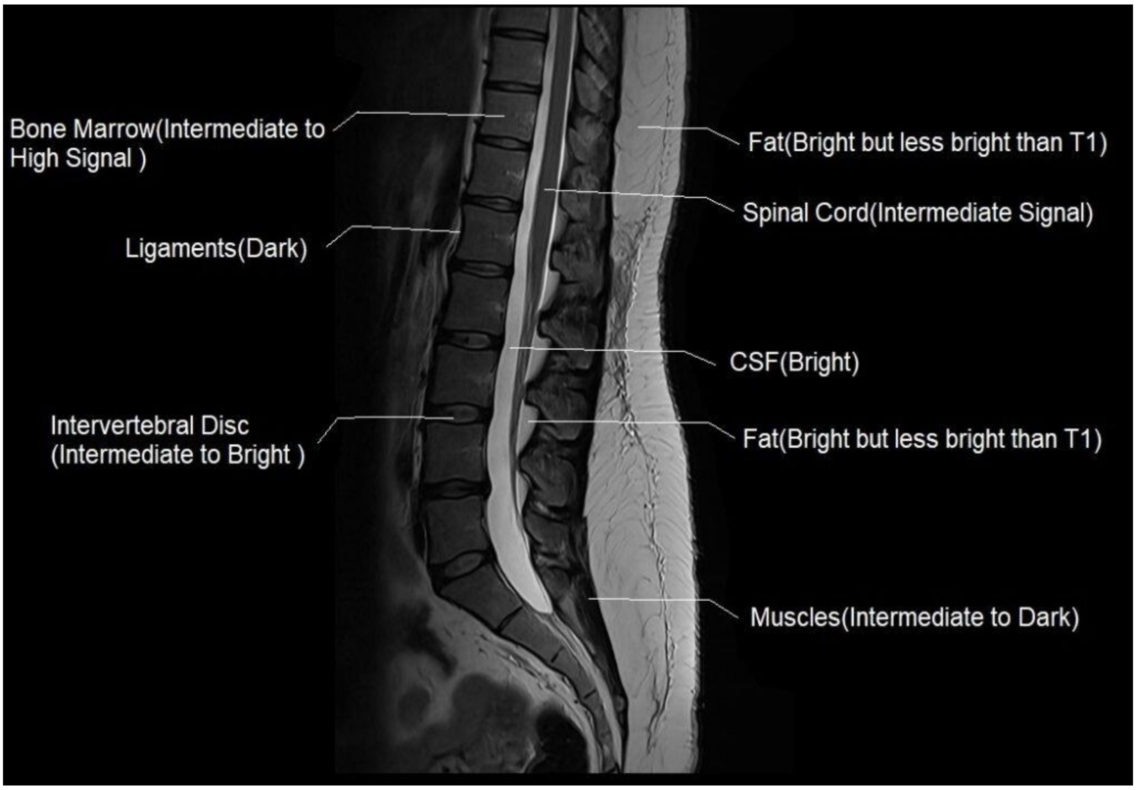


Fig 5: T2 MR Image of the lumbar spine (sagittal plane)

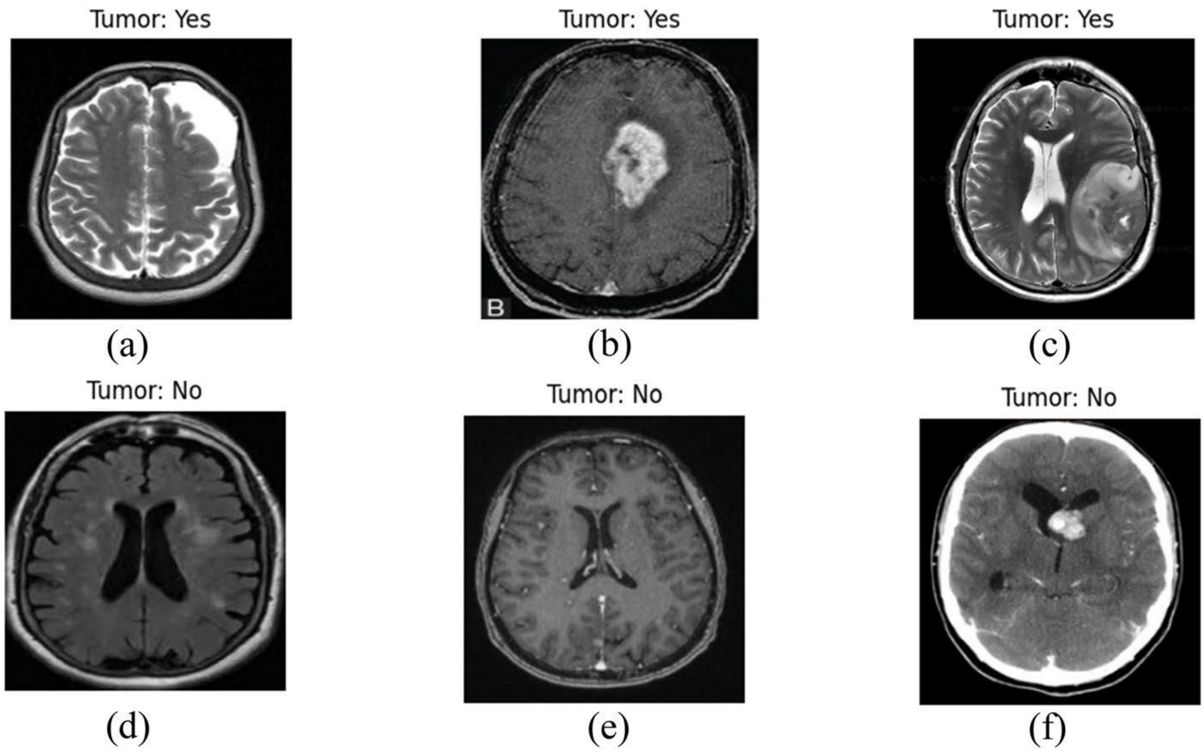


Fig 6: Comparison of Non-pathological and Pathological MR images
(with tumour/edema etc)

Participants shall analyse the given MRI dataset to understand its characteristics before Preprocessing.

The analysis should include:

- MRI resolution, contrast analysis.
- Comparison between Brain/Spine MRI submodalities in Dataset.
- Division of individual patient samples of Hackathon challenge dataset based on MRI sub-modalities (T1, T2, FLAIR, STIR).
- Discussion of dataset challenges relevant to MRI sub-modalities.
- Enumeration of datasets for training, testing and validation tasks

Expected Deliverables

- Dataset statistics of both training (Standard chosen Dataset), Testing/validation (Hackathon challenge Dataset).
- Image (MRI volumes) property assessment of the Datasets must be done with Contrast, complexity, Sharpness, Edge strength, Noise level, Mean, Deviation parameters.

STAGE 2 (MRI Dataset Pre-processing)

After assessing the properties of the MR images (MRI volumes in Dataset), preprocessing should be performed for denoising/ rescaling/artifact correction of the MRI samples and defocus the non-significant regions. Further basic MRI enhancement methods including Contrast enhancement, Histogram Equalization (HE)/Adaptive Histogram Equalization, /Contrast Limited Adaptive Histogram Equalization may be used for fundamental MRI enhancement. Once again, the image property assessment of the preprocessed MRI Dataset must be done, with Contrast, complexity, Sharpness, Edge strength, Noise level, Mean, Deviation parameters.

Further, the quality of the preprocessed/enhanced MR images (MRI volumes in Dataset) should be assessed and analysed using the following image quality evaluation metrics:

- Peak Signal-to-Noise Ratio (PSNR),
- Structural Similarity Index (SSIM),
- Mean Squared Error (MSE),
- Root Mean Square Error (RMSE),
- Universal Image Quality Index (UQI),
- Feature Similarity Index (FSIM),
- Gradient Magnitude Similarity Deviation (GMSD),
- Visual Information Fidelity (VIF),
- Blind/Reference less Image Spatial Quality Evaluator (BRISQUE),
- Natural Image Quality Evaluator (NIQE),
- Perception-based Image Quality Evaluator (PIQE), Entropy, and Learned Perceptual Image Patch Similarity (LPIPS).

Further Image property assessment (as followed in stage 1) must be done for pre-processed Datasets.

Expected Deliverables

- Justification of used preprocessing stage techniques relevant to Brain/Spine MRI Sub-modalities (T1, T2, FLAIR, STIR).
- Resizing, Scaling, denoising, artifact correction/reduction, Dataset augmentation.
- Annotation visualization of training Dataset.
- Curated datasets (training, testing/validation) for further AI based enhancement and Segmentation.

STAGE 3 (MR image quality enhancement of Pre-processed Dataset):

Based on the preprocessed MRI dataset obtained from Stage 2, the participating teams can apply either a Single Deep learning architecture or an Ensemble of Deep learning models/Transformer models/Self-supervised models to further enhance the quality of curated MRI dataset. The performance of the enhanced Brain MRI dataset should then be compared with the results reported in the research papers listed below (only for Brain MRI tasks) using the corresponding post-enhancement image quality evaluation metrics mentioned below.

References for research papers on enhancement of Brain/Spine MRI:

1. Ravi Kumar, Ashish Kumar Bhandari, "Spatial mutual information based detail preserving magnetic resonance image enhancement", Computers in Biology and Medicine 146 (2022) 105644.

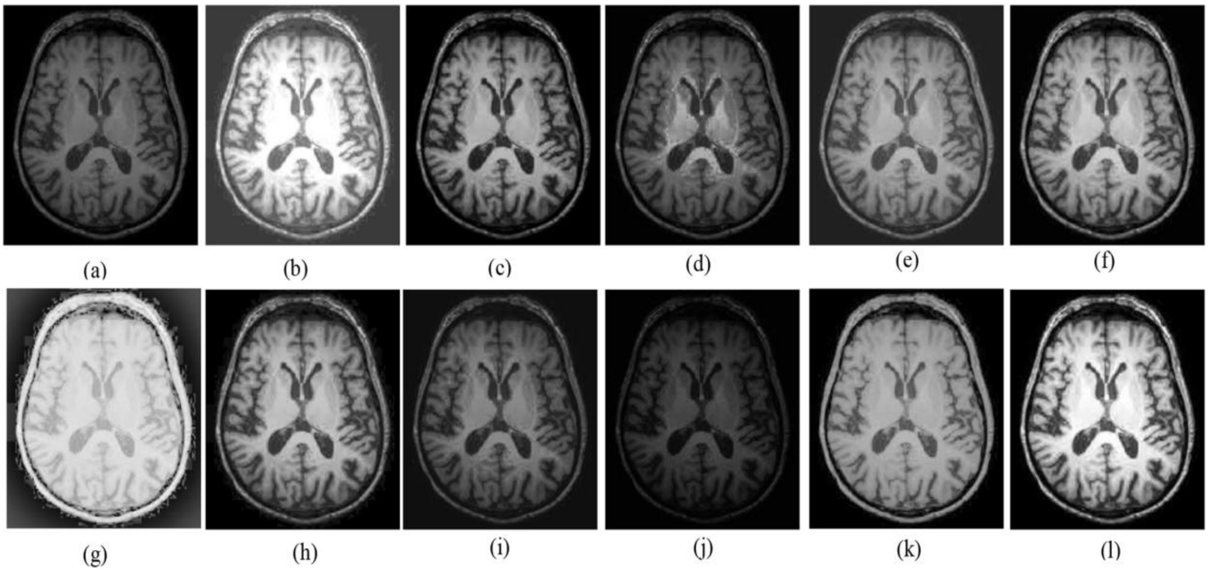


Fig1: Visual comparison of different enhancement methods for healthy input image
(a) Input image, (b) HE (c) WAHE (d) QDHE, (e) ROHIM (f) ESIHE (g) MSRCR (h) JHE (i) Fuzzy DCT (j) OHCICD (k) SIRE and (l) proposed method in above research paper

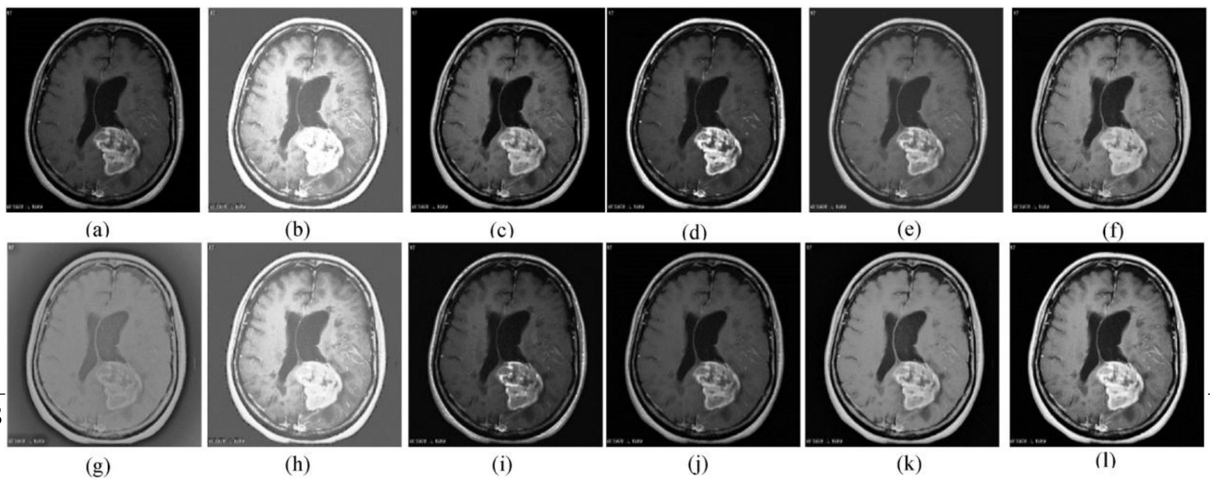


Fig 2: Visual comparison of different enhancement methods for pathological (diseased) input image
(a) Input image, (b) HE, (c) WAHE, (d) QDHE, (e) ROHIM, (f) ESIHE, (g) MSRC
(h) JHE (i) Fuzzy DCT (j) OHCICD (k) SIRE and (l) proposed method in above research paper.

2. Huayu Fan, Xiangyang Cao, et.al, "A New Method Used to Enhance the SPAIR Image of the Spine MRI", Current Medical Imaging, **1573-4056/24** DOI: 10.2174/0115734056251482231107072125, 2024.

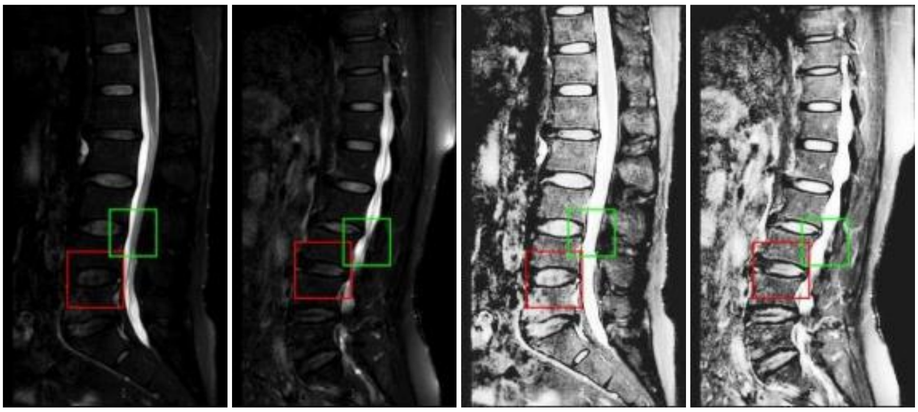


Fig 3: Different versions of LS Spine MRI enhancement (refer above research paper for details)

Further for both Brain/Spine MRI tasks, the comparison should be based on common quantitative metrics, including Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), Mean Squared Error (MSE), Root Mean Square Error (RMSE), Universal Image Quality Index (UQI), Feature Similarity Index (FSIM), Gradient Magnitude Similarity Deviation (GMSD), Visual Information Fidelity (VIF), Blind/Referenceless Image Spatial Quality Evaluator (BRISQUE), Natural Image Quality Evaluator (NIQE), Perception-based Image Quality Evaluator (PIQE), Entropy, and Learned Perceptual Image Patch Similarity (LPIPS). Further image property assessment (as followed in stage 1) must be done for enhanced datasets.

Expected Deliverables

- MRI enhancement evaluation for all curated MRI (Brain and Spine) sub-modality datasets with common quantitative metrics discussed above.
- Justification for the types of AI/Deep learning architecture/model used for MRI enhancement.
- Systematic evaluation of Training Loss, validation loss, learning curve, convergence epoch, overfitting gap, Cross validation accuracy for Deep Learning/AI model used.
- Systematic comparison study of AI/Deep learning models used for MRI enhancement based on Inference Latency, Throughput, GPU/CPU utilization, Memory Consumption, Model complexity.

STAGE 4 (Region of interest Segmentation in MRI dataset):

The enhanced MRI dataset from STAGE 3 is further need to be processed for region of interest segmentation. The region of interest will be CSF (cerebrospinal fluid), grey matter, white matter in healthy subjects of given MRI dataset, whereas for pathological (disease/disorder) subjects, the region of interest (ROI) will be tumour affected part of brain, edema affected region, lesions, metastasis region segmentation.

For LS (Lumbo sacral) spine datasets, ROI segmentation is to delineate Degenerative disc, Disc herniation, spinal stenosis. The common evaluation metrics for region of interest segmentation for both Brain/Spine MRI include Dice similarity co-efficient, Jaccard index, accuracy, sensitivity, specificity, precision, F1 score, Hausdorff Distance (HD), Average Surface Distance (ASD), Relative volume error, Grad-CAM, Attention maps. Further participants are also informed to evaluate the segmentation results with metrics relevant to mentioned in the below research papers.

Further for segmented portions (masks) of MRI dataset, Image quality assessment and Image property assessment must be done as described in Stage 3.

References for research papers on segmentation of pathological Brain MRI:
(Based on using external standard BRATS-CHALLENGE 2020 Dataset)

Theophraste Henry, et al, "Brain tumour segmentation with self-ensembled, deeply-supervised 3D U-net neural networks: a BraTS 2020 challenge solution.

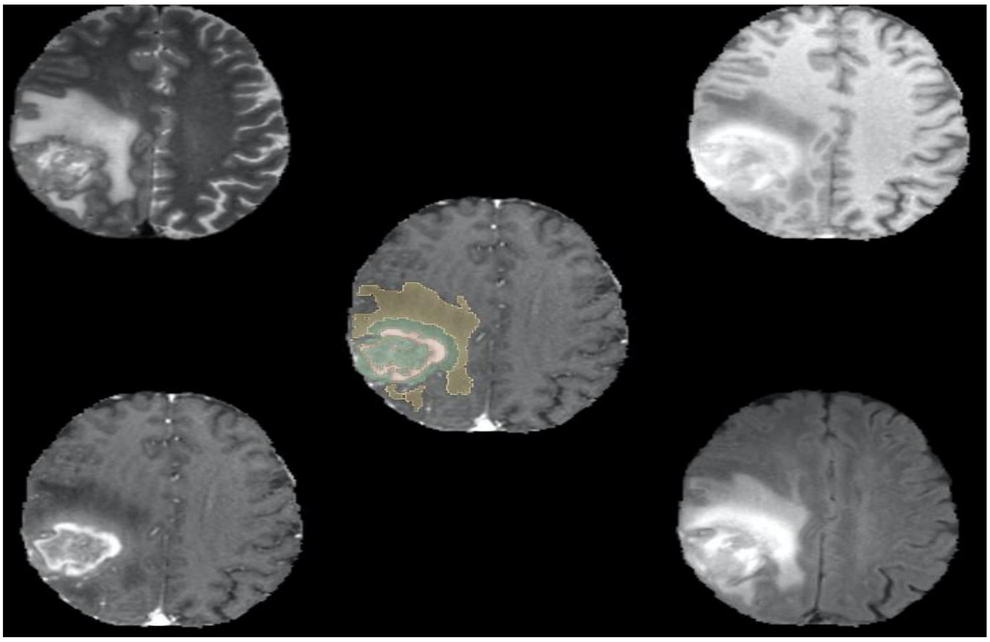


Fig. 1. Example of a brain tumor from the BraTS 2020 training dataset. Red: enhancing tumor (ET), Green: non enhancing tumor/ necrotic tumor (NET/NCR), Yellow:peritumoral edema (ED). Upper Left: T2 weighted sequence, Upper Right: T1 weighted sequence, Lower Left: T1-weighted contrast enhanced sequence, Lower Right:

FLAIR sequence Middle: T1-weighted contrast enhanced sequence with labelmap overlay. (Research paper results only)

References for Lumbar Spine intervertebral discs segmentation:

(NOTE: Participants must use given Hackathon Dataset for Spine MRI related tasks)

1. Zhao, D., Qin, R., Chai, Z. *et al.* Spinal disease image segmentation technology integrating U-ResNet and shape-aware attention. *Sci Rep* **16**, 12465 (2026). https://doi.org/10.1038/s41598-026-42870-9

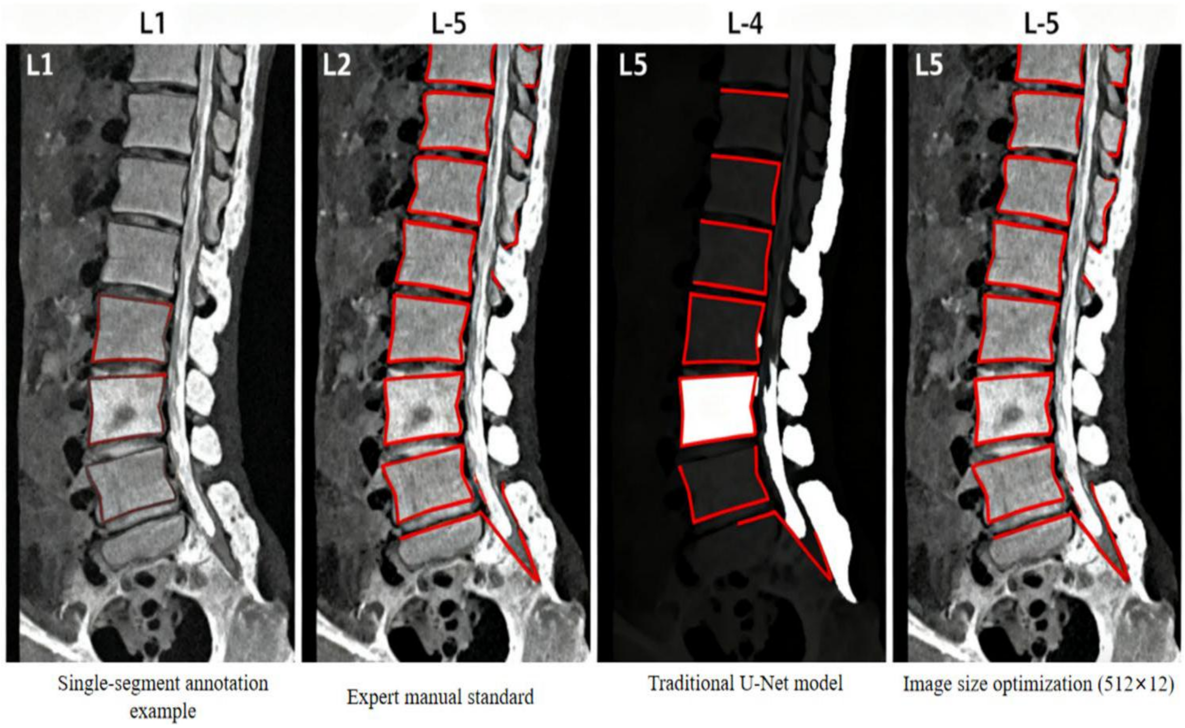


Fig 1.1: Comparison of effects of different Segmentation methods on multiple (LS) lumbar segments. (research paper results only)

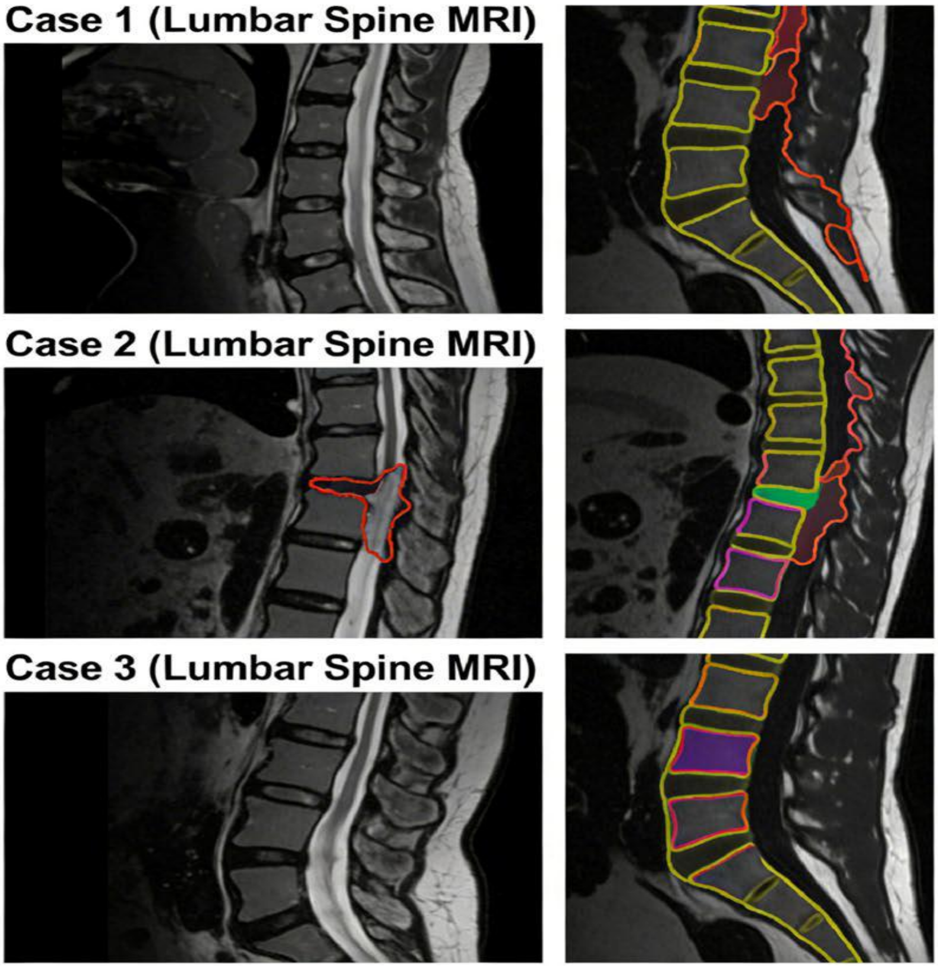


Fig 1.2: Qualitative segmentation results of 3 different cases, the above figure includes original images, model segmentation results. (Research paper results only)

2. Jianlong Cai1, et.al, "Advancing Fine-Grained Spine Segmentation Through Visual-Language Model with Omni- and Pixel-Level Semantic Enhancements", 8th Chinese Conference, PRCV 2025, Shanghai, China, October 15–18, 2025, Proceedings, Part XIV.

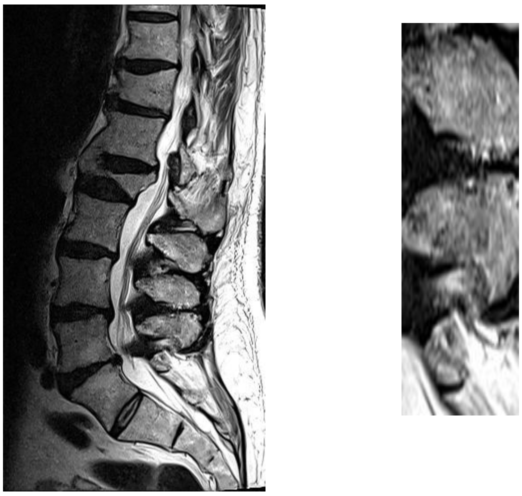


Fig 2.1: Original image from dataset

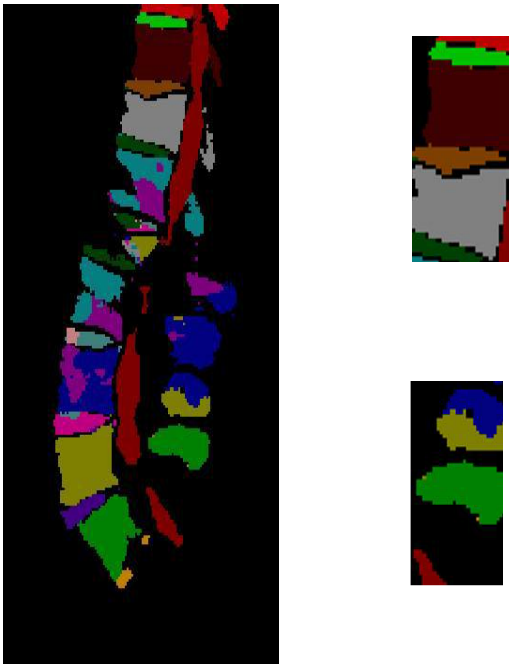


Fig 2.2: Segmentation results using U-net model (research paper results)

Expected Deliverables

- Common evaluation metrics-based MRI ROI segmentation for all MRI (Brain and Spine) sub-modality (testing/validation) datasets.
- Justification for the AI/Deep learning model used for MRI ROI segmentation.
- Systematic evaluation of Training Loss, validation loss, learning curve, convergence epoch, overfitting gap, Cross validation accuracy for Deep Learning/AI model used.
- Comparison of AI/Deep learning models used for MRI ROI segmentation based on Inference Latency, Throughput, GPU/CPU utilization, Memory Consumption, Model complexity.

Expected Hackathon Final Deliverables

Each participating team shall submit:

- Source code (ZIP format)
- Trained/tested AI/DL model architecture
- MRI Enhancement and segmentation results (COCO JSON format)
- Technical report (3–5 pages)
- Presentation slides

Evaluation Criteria

Criteria	**Weightage**
Dataset Analysis	20%
Dataset Preparation & Preprocessing	10%
Design of MRI Enhancement model	30%
Design of MRI ROI-based Segmentation model	30%
Technical Report & Presentation	10%
Total	100%

Hardware and Software Requirements

Hardware Requirements (any of these)

- Minimum: Intel Core i5 (or equivalent), 12 GB RAM, NVIDIA RTX 3050 (or equivalent GPU with at least 8 GB GPU memory).
- Recommended: Intel Core i7/Ryzen 7 (or higher), 32 GB RAM, NVIDIA RTX 4060 or above (or equivalent GPU with 12–16 GB GPU memory).

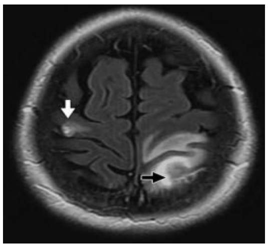
- Cloud Computing Platforms: Participants may also use any of the following GPU-enabled cloud platforms:
  - Google Colab (Free, Pro, or Pro+)
  - Kaggle Notebooks (GPU)
  - Institutional GPU servers
  - Other cloud platforms supporting CUDA-enabled GPUs

Software Requirements

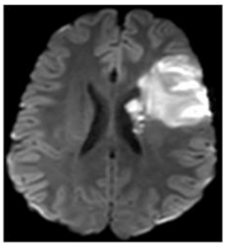
- MATLAB 2023a/2024a/2025a
- Python 3.10 or later
- PyTorch
- OpenCV
- NumPy
- Torchvision

APPENDIX

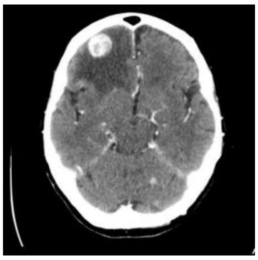
Reference sample images for Brain MRI pathological cases:



Edema



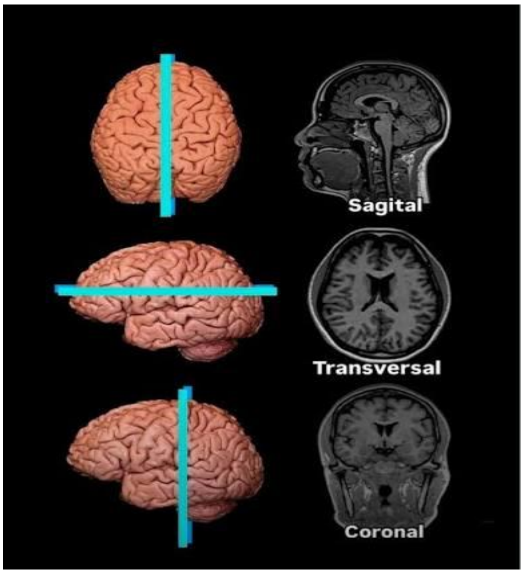
Stroke/infarction



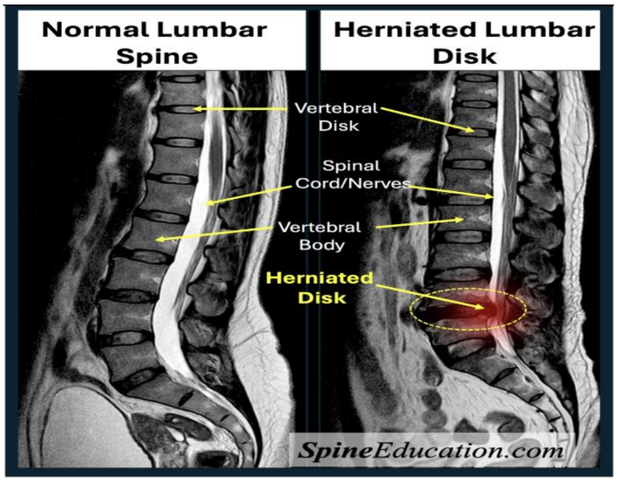
Lesion

Multiple Sclerosis

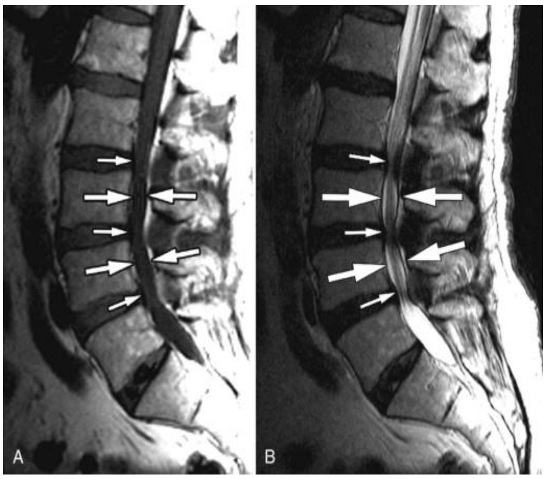
MRI Acquisition planes:



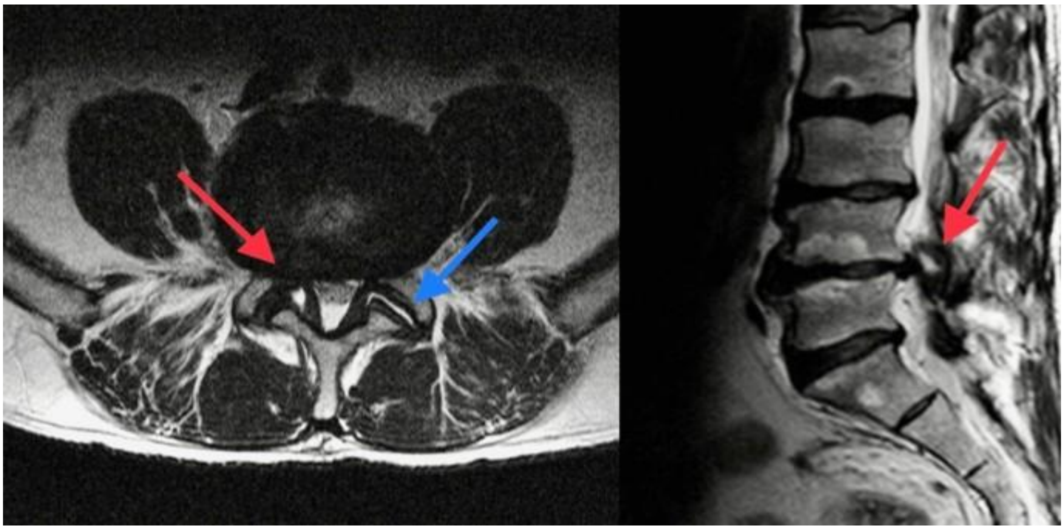
Reference sample images for Spine MRI pathological cases:



Disc herniation



Spinal stenosis



Degenerative LS Spine disc